

Reviewer Pack — Minimal Rank of Multivariate Polynomials in Boolean Function...

Entry 98a6133a0fd0 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-26 22:34:46 UTC

Minimal Rank of Multivariate Polynomials in Boolean Functions vs XOR-AND Tree Width

Entry ID: 98a6133a0fd0 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 22:34:46 UTC

1. Conjecture

Field A (mathematical branch): Multivariate Polynomial Theory **Field B** (complexity object): Complexity Theory: XOR-AND Tree Width

Statement:

{'concrete_statement': 'For any Boolean function f with n variables, the minimal rank of the multivariate polynomial representing f is at most $c * \log(n) / \log(\log(n))$ for some constant c .', 'falsifier_statement': 'There exists a Boolean function f with n variables such that the minimal rank of the multivariate polynomial representing f exceeds $c * \log(n) / \log(\log(n))$ for any constant c .')}

Rationale (proposer's reasoning):

{'reason_1': 'Multivariate polynomials can be used to represent Boolean functions, and their minimal rank is a measure of complexity.', 'reason_2': 'The XOR-AND tree width is a measure of the complexity of a Boolean function, and it has been shown that there are connections between polynomial degree and tree width.', 'reason_3': 'Exploring this connection could provide new insights into the complexity of Boolean functions.'}

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 81d563f063f3d0eb

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The average minimal rank of multivariate polynomials representing randomly generated Boolean functions with n variables is at most $c * \log(n) / \log(\log(n))$ over 30 seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.90	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def polynomial_rank(poly):
        n = len(poly)
        if n == 1:
            return 1
        rank = 0
        for i in range(1, n):
            if poly[i] != 0:
                rank += 1
        return rank

    def min_polynomial_rank(f):
        n = int(math.log2(len(f)))
        min_rank = float('inf')
        for _ in range(30): # Sample multiple polynomials to get a good estimate
            poly = generate_boolean_function(n)
            rank = polynomial_rank(poly)
            if rank < min_rank:
                min_rank = rank
        return min_rank

    n = random.randint(5, 40) # Sweep n through at least 4 distinct sizes
    f = generate_boolean_function(n)
    min_rank = min_polynomial_rank(f)

    metric_name = "minimal_rank"
    metric_value = min_rank / (math.log(n) / math.log(math.log(n)))
    instances_tested = 30
    conjecture_holds = metric_value <= 10 * (math.log(n) / math.log(math.log(n))) # Use a non-trivial
```

```

counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": metric_name,
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if len(sys.argv) > 1 else [2, 3, 5, 7, 11, 13, 17, 19,

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        for r in results:
            if not r["conjecture_holds"]:
                counterexample = r["counterexample"]
                first_failing_seed = seed
                break
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot confirm whether the conjecture is supported or falsified. | next: Re-run the test with increased time limits to ensure it completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17268
2	propose	ollama_remote	glm4:latest	0	0	11097
3	preregistration	ollama_remote	glm4:latest	0	0	5240
4	novelty	ollama_remote	glm4:latest	0	0	4728
5	novelty	ollama_remote	glm4:latest	0	0	5258
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13643
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8287
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6919
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8622
10	judge	ollama_remote	glm4:latest	0	0	12408

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 93470 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/98a6133a0fd0.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/98a6133a0fd0.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/98a6133a0fd0.tar.gz* (if generated)